Groundwater

Katie Markovich, EPS 522/ ENVS 423L (Fall 2025)

Housekeeping

- Guest lecture next Tuesday
- HW7 assigned next Tuesday
- No class November 11

522 Term Project

Tech Memo:

- Word template to be posted soon
- Keep it concise!
- Each group member will submit same document to Canvas

Presentation:

- 15-minutes
- Brief background, data used, water balance result, challenges faced.
- Homework #9 will be a peer evaluation of the presentations (I will hand out forms on the day).

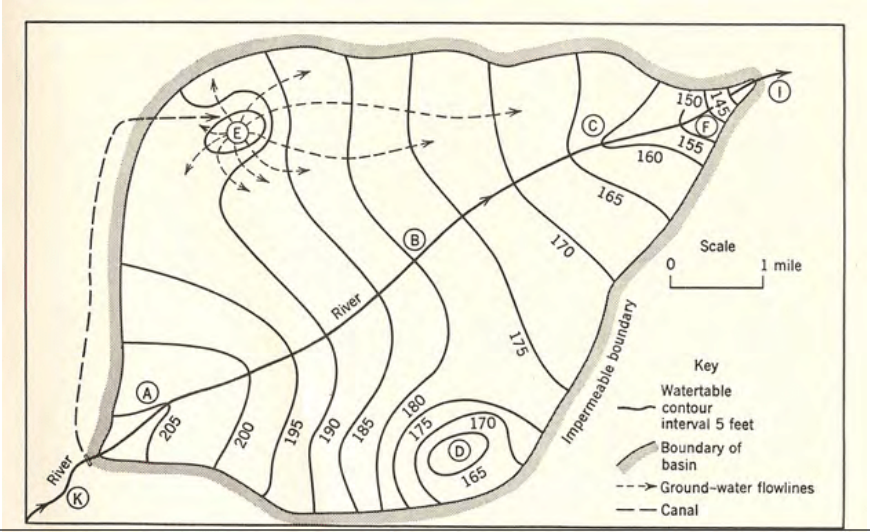
Rubrics for tech memo and presentation will also be posted to Canvas soon.

Learning Objectives

1. Aquifer Types and Properties
2. Groundwater Flow Equation(s)
3. Regional Groundwater Flow
4. GW/SW Interactions
5. Groundwater Balance Components
6. Groundwater Development

Head Contour Map

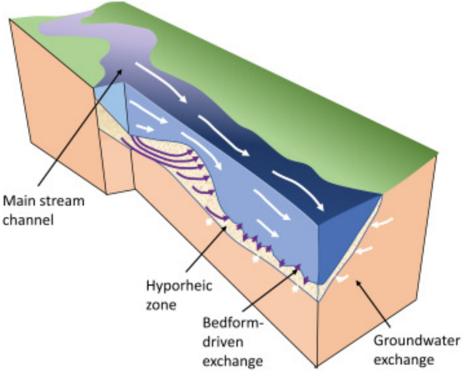
Not just for flow nets!



Hyporheic Zone

Hyporheic zone: porous media immediately adjacent to stream channel where a high degree of interaction between SW/GW occurs.

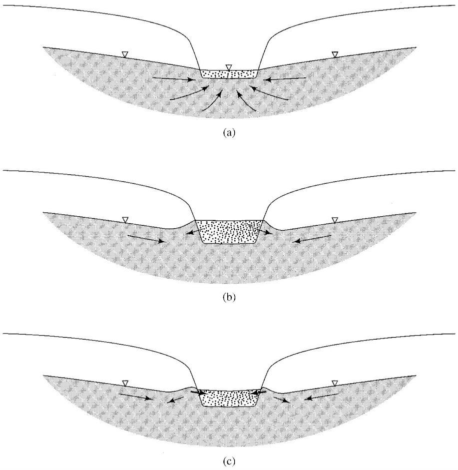
This zone may be small in area/volume, but plays an outsized influence in aquatic habitat, temperature regulation, solute attenuation, and biogeochemical processes.



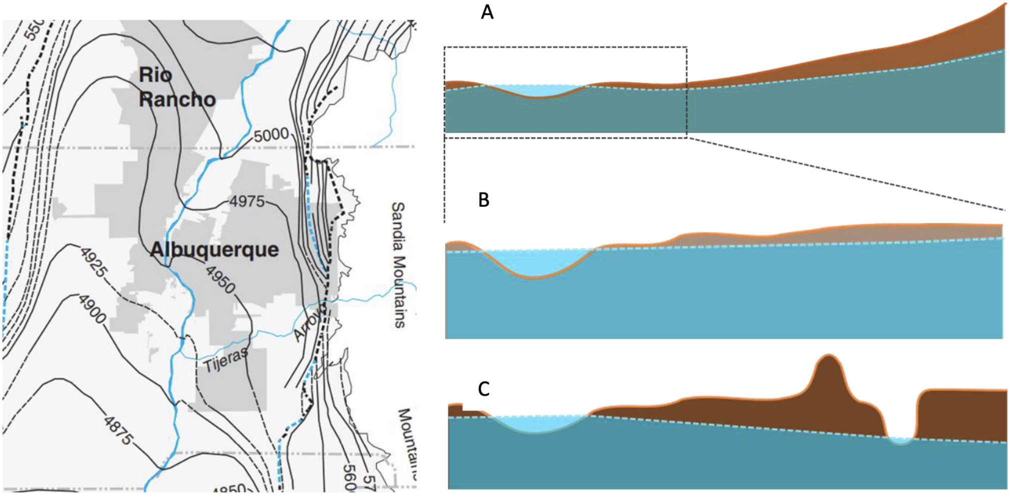
Bank Storage

Bank storage: leading edge of flood wave produces higher stream stage than water level in the channel bank, inducing flow from stream to bank. After this wave passes, that gradient is reversed the saturated bank wedge drains back to the stream.

Important for flood wave mitigation, as well as a key source of water for riparian vegetation.



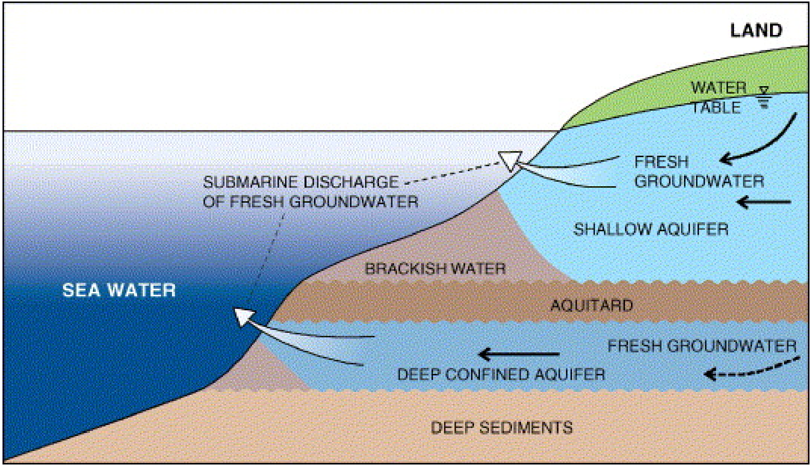
Rio Grande Bank Storage



Heller (2018) (UNM Water Resources Professional Project)

Flow to Oceans

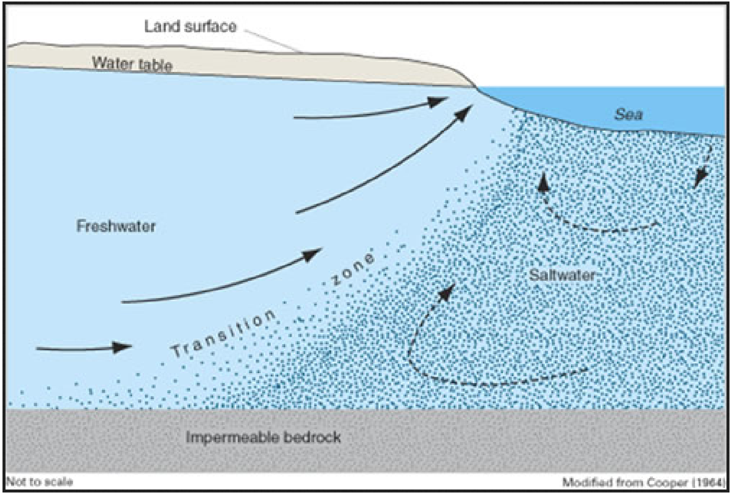
Groundwater that does not contribute to streamflow, lakes/wetlands, or playa evaporation (in "closed" basins), discharges to the ocean as **submarine groundwater discharge.**



Burnett et al. (2006)

Flow to Oceans

Due to its lower density, freshwater remains above seawater, with a transition zone of varying thickness.



Flow to Oceans

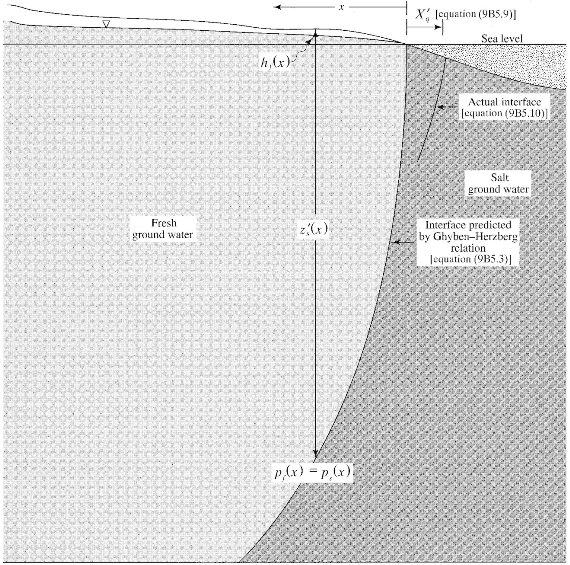
Fresh, brackish, and saline water are distinguished by total dissolved solids (TDS) having units of mg/L (or parts per million [ppm]).

Type	TDS (mg/L)
Drinking water	≤ 500
Fresh water	<1000
Mildly brackish water	1000–5000
Moderately brackish water	5000–15,000
Heavily brackish water	15,000–35,000
Seawater	$\geq 35{,}000$

Source: Watson et al.[14]

In the case of seawater, chloride makes up more than 50% of the TDS.

Ghyben-Herzberg Relation



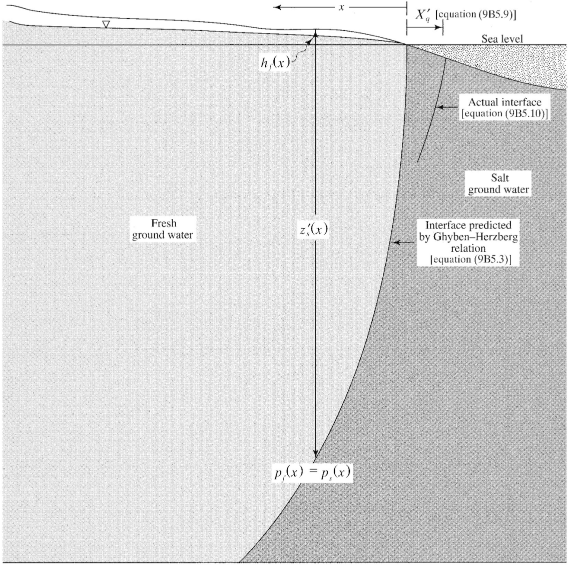
For a homogeneous, unconfined aquifer under steady-state conditions we can assume a sharp interface between fresh and saline water.

Ghyben-Herzberg Relation

$$p_f(x) = \gamma_f \cdot [h_f(x) + z_s(x)]$$

$$p_s(x) = \gamma_s \cdot z_s(x)$$

where p is pressure and γ is weight density of fresh and saline water, h_f(x) is the elevation above sea level, and z is elevation below sea level of the interface.



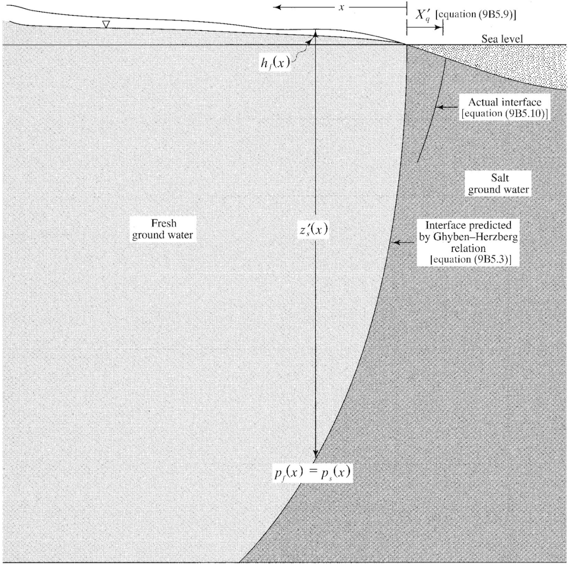
Ghyben-Herzberg Relation

Assuming hydrostatic equilibrium,
$(p_f(x) = p_s(x))$:

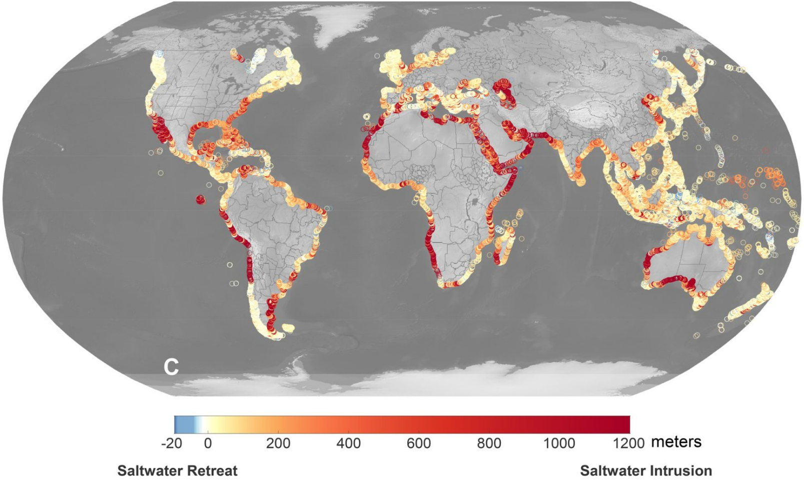
$$z_s(x) = \Gamma \cdot h_f(x)$$

$$\Gamma = \frac{\gamma_f}{\gamma_s - \gamma_f} = \frac{9800}{10,045 - 9800} = 40$$

or the fresh-salt interface is at a depth below sea level that is 40 times the height of the WT above sea level.



Seawater Intrusion in 2100



Adams et al. (2024)

Causes?

- pumping
- sea level rise
- decreased recharge

Groundwater Balance

Regional WB, under steady-state, pre-development (no pumping) conditions:

$$P + G_{in} = Q + ET + G_{out}$$

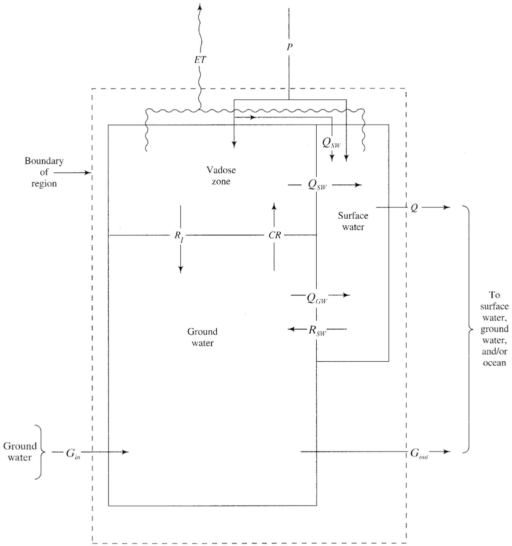
Groundwater Balance:

$$R_i + R_{sw} + G_{in} = CR + Q_{gw} + G_{out}$$

Net recharge:

$$R_n = R_i + R_{sw} - CR$$

where CR is capillary rise.



Quantifying Groundwater Balance Components

1. Recharge from infiltration (Diffuse Recharge)
   - Water Balance (P minus ET)
   - Water Level Fluctuation
   - Inverse Methods
   - Chloride Mass Balance
2. Recharge from surface water (Focused Recharge)
   - Direct Measurement
   - Differential Gaging
   - Temperature
3. Capillary rise
   - No one cares about this
4. Deep seepage (G_{in} and G_{out})
   - Head Contours
   - Water Balance

Diffuse Recharge: Water Balance

Assuming no change in storage and balanced groundwater inflows/outflows:

$$R = P - AET - Q$$

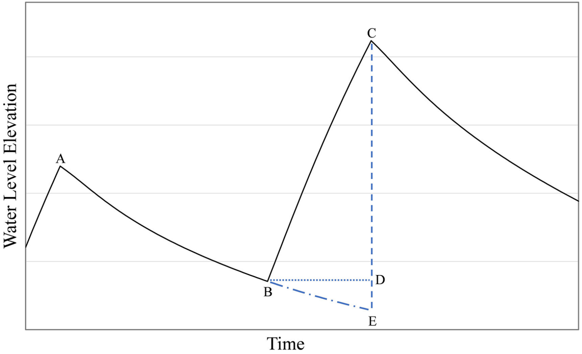
Limitations:

- AET is really hard to quantify
- It relies on good quantifications of P + Q

Diffuse Recharge: Water Level Fluctuation

The WTF method assumes that a rise in water level occurs as a result of recharge across the water table.

$$R = S_y \cdot \frac{\Delta h}{\Delta t}$$



Becke et al., (2024)

Limitations:

- negates other influences (tides, air pressure, pumping)
- extrapolating B-E
- S_y is uncertain
- localized

Diffuse Recharge: Inverse Methods

For 3-D groundwater flow, the variable we solve for is head, h(x,y,z,t), assuming we know or can guess the initial distribution of heads, boundary conditions, K, S_s, and R.

$$\frac{\partial}{\partial x}K_x(\frac{\partial h}{\partial x}) + \frac{\partial}{\partial y}K_y(\frac{\partial h}{\partial y}) + \frac{\partial}{\partial z}K_z(\frac{\partial h}{\partial z}) = S_s\frac{\partial h}{\partial t} + R$$

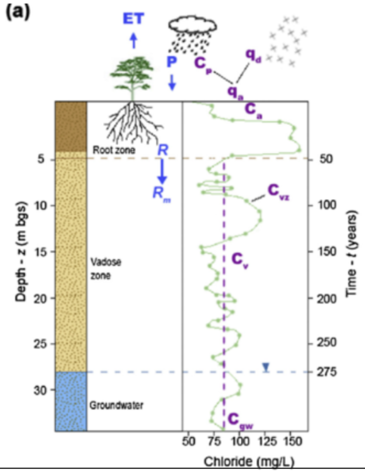
An inverse approach to estimating recharge is essentially adjusting values of K, S_s, and R to minimize the difference between *modeled* and *observed* heads.

Limitations:

- ill-posed inverse problem (equifinality)
- computationally intensive

Diffuse Recharge: Chloride Mass Balance

Assumes all chloride in groundwater is derived from precipitation and that chloride concentration occurs by evaporation prior to recharge.

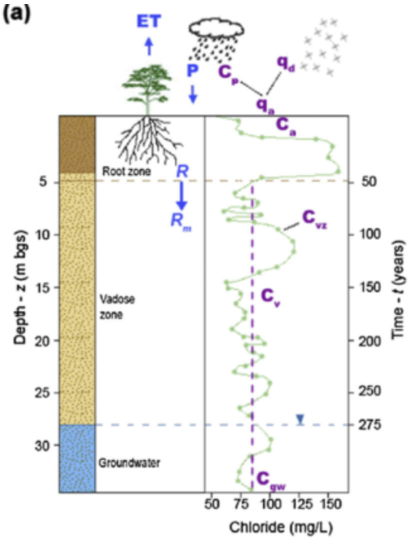


Manna et al. (2017)

$$R = \frac{P \cdot C_p^*}{C_{gw}}$$

where P is long-term average precip, C_p^* is the weighted average chloride concentration of precip, and C_{gw} is the chloride concentration in groundwater.

Diffuse Recharge: Chloride Mass Balance



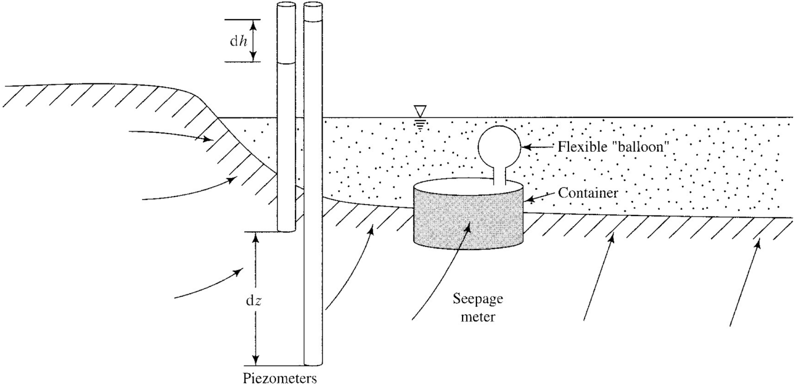
Manna et al. (2017)

Assumptions:

1. no other sources of chloride to GW
2. chloride is conservative
3. long-term precip rates and chloride concentrations are stable
4. no surface runoff leaving the aquifer (precip either evaporated or recharged)
5. no evap of GW upgradient of sampling pt

Focused Recharge: Direct Measurement

Paired measurements of stream stage and groundwater head.

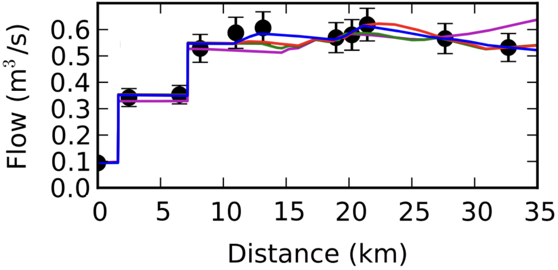


Limitations:

- warible process, these are just point measurements

Focused Recharge: Differential Gaging

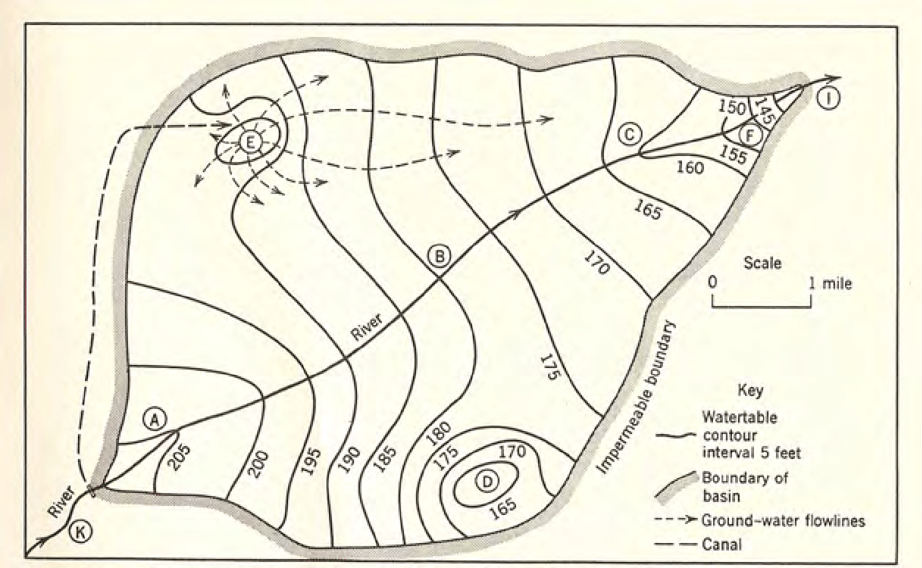
If surface runoff is negligible, the net groundwater inflow/outflow to streams is the difference between stream discharge at an upstream and downstream gaging location.



Limitations:

- no evap assumption
- seepage needs to be > measurement error
- snapshot in time / labor-intensive

Deep Seepage: Head Contours



Deep Seepage: Water Balance

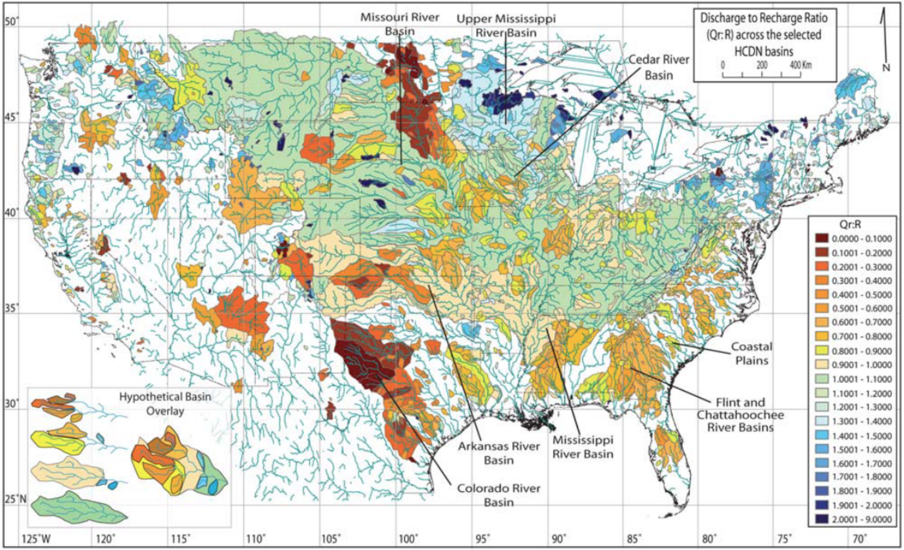
If there are no groundwater inflows/outflows (or they perfectly balance each other) and there are no long-term trends in storage, the water balance equation simplifies to:

$$P - ET = R = Q$$

If R $\neq$ Q (and P and ET are reasonably quantified), then the difference implies deep seepage.

Schaller et al. (2009) used the ratio of Q to R to characterize whether a watershed is an "importer" (Q/R $>$ 1) or an "exporter" (Q/R $<$ 1) of groundwater.

Watershed Importers/Exporters



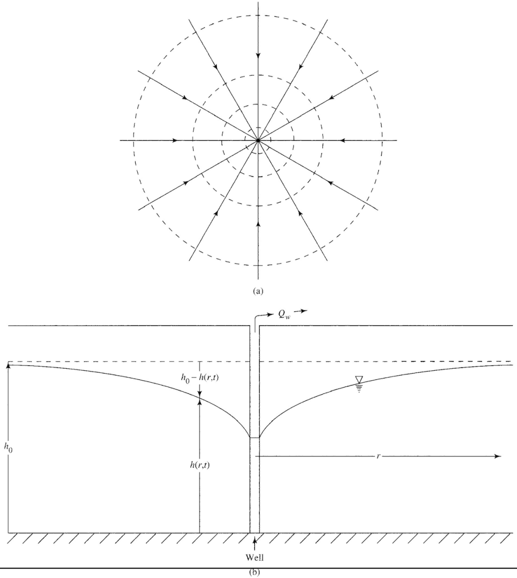
Groundwater Development

Groundwater development is the process of accessing and utilizing groundwater by drilling and pumping wells.

Pumping Terminology

- **Drawdown**: Lowering of head due to pumping. Largest closest to the well.

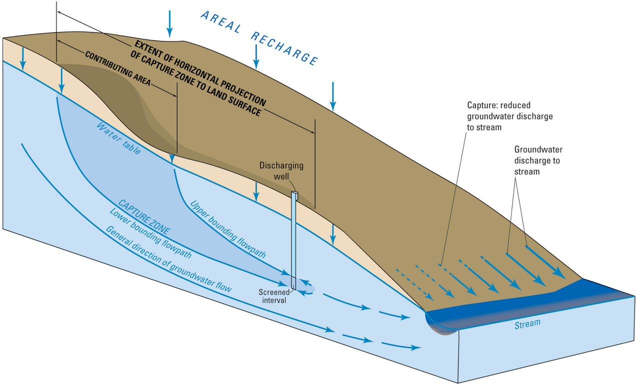
- **Cone of Depression**: 3-D expression of drawdown near a well.



Pumping Terminology

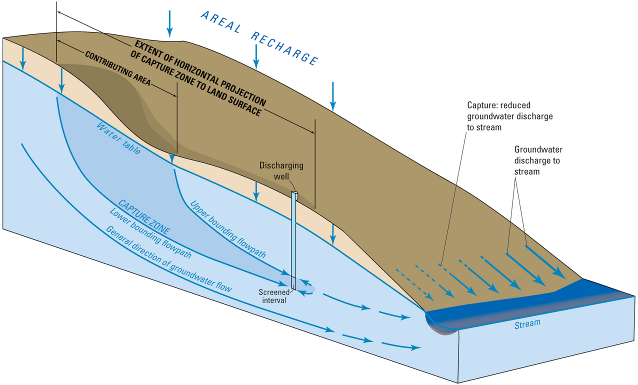
- **Capture Zone**: 3-D volumetric portion of a groundwater-flow field that discharges water to a well.

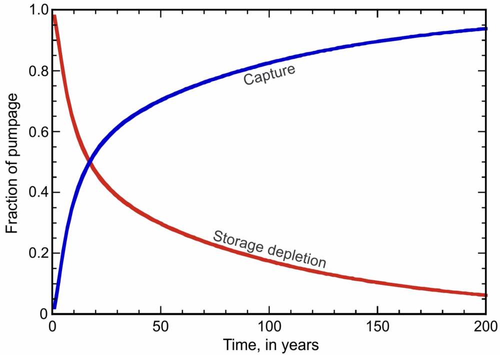
- **Contributing Area**: 2-D areal extent of that portion of the capture zone.



Capture

Capture: pumping-induced increases in aquifer recharge, decreases in aquifer discharge, or a combination of the two.





"All water discharged by wells is balanced by a loss of water somewhere." (Theis, 1940)

Radial Flow to a Well

Fully penetrating well in a confined, homogeneous aquifer of infinite extent, drawdown (s) can be calculated as:

$$s = \frac{Q}{4\pi T} \cdot W(u)$$

$$u = \frac{r^2 \cdot S}{4Tt}$$

$$W(u) = -0.577216 - \ln(u) + u - \frac{u^2}{2 \cdot 2!} + \frac{u^3}{3 \cdot 3!} - ...$$

- s = drawdown (L)
- Q = pumping rate (L3/T)
- T = transmissivity (K * b, L^2/T)
- W(u) = well function
- r = distance from well (L)
- S = storativity (-)
- t = time elapsed since pumping began

Radial Flow to a Well

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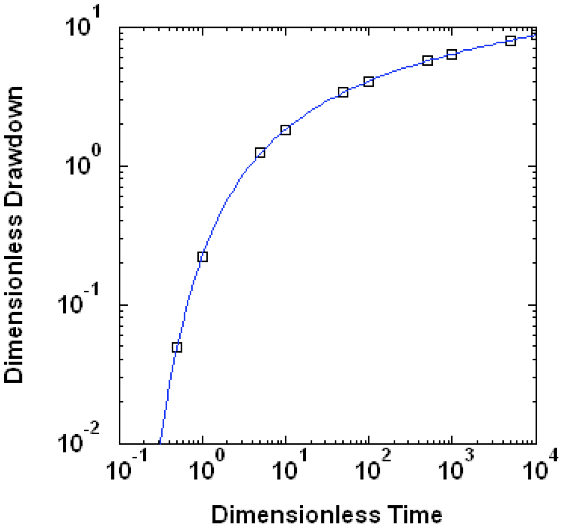
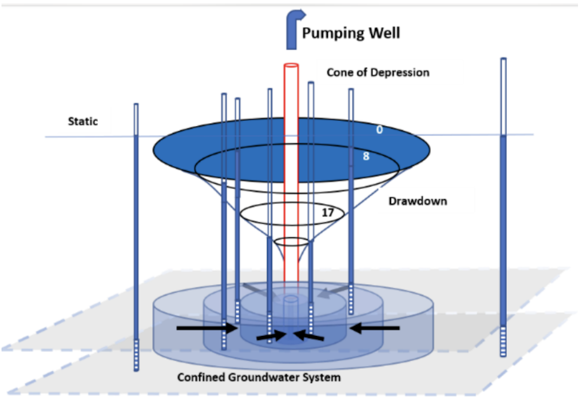
$$W(u) = -0.577216 - \ln(u) + u - \frac{u^2}{2 \cdot 2!} + \frac{u^3}{3 \cdot 3!} - .$$

TABLE 2.—*Values of* W(u) *for values of* u *between* 10^{-15} *and 9.9*

N \ u	$N\times10^{-15}$	$N\times10^{-14}$	$N\times10^{-13}$	$N\times10^{-12}$	$N\times10^{-11}$	$N\times10^{-10}$	$N\times10^{-9}$	$N\times10^{-8}$	$N\times10^{-7}$	$N\times10^{-6}$	$N\times10^{-5}$	$N\times10^{-4}$	$N\times10^{-3}$	$N\times10^{-2}$	$N\times10^{-1}$	N
1.0	33.9616	31.6590	29.3564	27.0538	24.7512	22.4486	20.1460	17.8435	15.5409	13.2383	10.9357	8.6332	6.3315	4.0379	1.8229	0.2194
1.1	33.8662	31.5637	29.2611	26.9585	24.6559	22.3533	20.0507	17.7482	15.4456	13.1430	10.8404	8.5379	6.2363	3.9436	1.7371	.1860
1.2	33.7792	31.4767	29.1741	26.8715	24.5689	22.2663	19.9637	17.6611	15.3586	13.0560	10.7534	8.4509	6.1494	3.8576	1.6595	.1584
1.3	33.6992	31.3966	29.0940	26.7914	24.4889	22.1863	19.8837	17.5811	15.2785	12.9759	10.6734	8.3709	6.0695	3.7785	1.5889	.1355
1.4	33.6251	31.3225	29.0199	26.7173	24.4147	22.1122	19.8096	17.5070	15.2044	12.9018	10.5993	8.2968	5.9955	3.7054	1.5241	.1162
1.5	33.5561	31.2535	28.9509	26.6483	24.3458	22.0432	19.7406	17.4380	15.1354	12.8328	10.5303	8.2278	5.9266	3.6374	1.4645	.1000
1.6	33.4916	31.1890	28.8864	26.5838	24.2812	21.9786	19.6760	17.3735	15.0709	12.7683	10.4657	8.1634	5.8621	3.5739	1.4092	.08631
1.7	33.4309	31.1283	28.8258	26.5232	24.2206	21.9180	19.6154	17.3128	15.0103	12.7077	10.4051	8.1027	5.8016	3.5143	1.3578	.07465
1.8	33.3738	31.0712	28.7686	26.4660	24.1634	21.8608	19.5583	17.2557	14.9531	12.6505	10.3479	8.0455	5.7446	3.4581	1.3089	.06471
1.9	33.3197	31.0171	28.7145	26.4119	24.1094	21.8068	19.5042	17.2016	14.8990	12.5964	10.2939	7.9915	5.6906	3.4050	1.2649	.05620
2.0	33.2684	30.9658	28.6632	26.3607	24.0581	21.7555	19.4529	17.1503	14.8477	12.5451	10.2426	7.9402	5.6394	3.3547	1.2227	.04890
2.1	33.2196	30.9170	28.6145	26.3119	24.0093	21.7067	19.4041	17.1015	14.7989	12.4964	10.1938	7.8914	5.5907	3.3069	1.1829	.04261
2.2	33.1731	30.8705	28.5679	26.2653	23.9628	21.6602	19.3576	17.0550	14.7524	12.4498	10.1473	7.8449	5.5443	3.2614	1.1454	.03719
2.3	33.1286	30.8261	28.5235	26.2209	23.9183	21.6157	19.3131	17.0106	14.7080	12.4054	10.1028	7.8004	5.4999	3.2179	1.1099	.03250
2.4	33.0861	30.7835	28.4809	26.1783	23.8758	21.5732	19.2706	16.9680	14.6654	12.3628	10.0603	7.7579	5.4575	3.1763	1.0762	.02844
2.5	33.0453	30.7427	28.4401	26.1375	23.8349	21.5323	19.2298	16.9272	14.6246	12.3220	10.0194	7.7172	5.4167	3.1365	1.0443	.02491
2.6	33.0060	30.7035	28.4009	26.0983	23.7957	21.4931	19.1905	16.8880	14.5854	12.2828	9.9802	7.6779	5.3776	3.0983	1.0139	.02185
2.7	32.9683	30.6657	28.3631	26.0606	23.7580	21.4554	19.1528	16.8502	14.5476	12.2450	9.9425	7.6401	5.3400	3.0615	.9849	.01918
2.8	32.9319	30.6294	28.3268	26.0242	23.7216	21.4190	19.1164	16.8138	14.5113	12.2087	9.9061	7.6038	5.3037	3.0261	.9573	.01686
2.9	32.8968	30.5943	28.2917	25.9891	23.6865	21.3839	19.0813	16.7788	14.4762	12.1736	9.8710	7.5687	5.2687	2.9920	.9309	.01482
3.0	32.8629	30.5604	28.2578	25.9552	23.6526	21.3500	19.0474	16.7449	14.4423	12.1397	9.8371	7.5348	5.2349	2.9591	.9057	.01305
3.1	32.8302	30.5276	28.2250	25.9224	23.6198	21.3172	19.0146	16.7121	14.4095	12.1069	9.8043	7.5020	5.2022	2.9273	.8815	.01149
3.2	32.7984	30.4958	28.1932	25.8907	23.5880	21.2855	18.9829	16.6803	14.3777	12.0751	9.7726	7.4703	5.1706	2.8965	.8583	.01013
3.3	32.7676	30.4651	28.1625	25.8599	23.5573	21.2547	18.9521	16.6495	14.3470	12.0444	9.7418	7.4395	5.1399	2.8668	.8361	.008939
3.4	32.7378	30.4352	28.1326	25.8300	23.5274	21.2249	18.9223	16.6197	14.3171	12.0145	9.7120	7.4097	5.1102	2.8379	.8147	.007891
3.5	32.7088	30.4062	28.1036	25.8010	23.4985	21.1959	18.8933	16.5907	14.2881	11.9855	9.6830	7.3807	5.0813	2.8099	.7942	.006970
3.6	32.6806	30.3780	28.0755	25.7729	23.4703	21.1677	18.8651	16.5625	14.2599	11.9574	9.6548	7.3526	5.0532	2.7827	.7745	.006160
3.7	32.6532	30.3506	28.0481	25.7455	23.4429	21.1403	18.8377	16.5351	14.2325	11.9300	9.6274	7.3252	5.0259	2.7563	.7554	.005448
3.8	32.6266	30.3240	28.0214	25.7188	23.4162	21.1136	18.8110	16.5085	14.2059	11.9033	9.6007	7.2985	4.9993	2.7306	.7371	.004820
3.9	32.6006	30.2980	27.9954	25.6928	23.3902	21.0877	18.7851	16.4825	14.1799	11.8773	9.5748	7.2725	4.9735	2.7056	.7194	.004267
4.0	32.5753	30.2727	27.9701	25.6675	23.3649	21.0623	18.7598	16.4572	14.1546	11.8520	9.5495	7.2472	4.9482	2.6813	.7024	.003779
4.1	32.5506	30.2480	27.9454	25.6428	23.3402	21.0376	18.7351	16.4325	14.1299	11.8273	9.5248	7.2225	4.9236	2.6576	.6859	.003349
4.2	32.5265	30.2239	27.9213	25.6187	23.3161	21.0136	18.7110	16.4084	14.1058	11.8032	9.5007	7.1985	4.8997	2.6344	.6700	.002969
4.3	32.5029	30.2004	27.8978	25.5952	23.2926	20.9900	18.6874	16.3884	14.0823	11.7797	9.4771	7.1749	4.8762	2.6119	.6546	.002633
4.4	32.4800	30.1774	27.8748	25.5722	23.2696	20.9670	18.6644	16.3619	14.0593	11.7567	9.4541	7.1520	4.8533	2.5899	.6397	.002336
4.5	32.4575	30.1549	27.8523	25.5497	23.2471	20.9446	18.6420	16.3394	14.0368	11.7342	9.4317	7.1295	4.8310	2.5684	.6253	.002073
4.6	32.4355	30.1329	27.8303	25.5277	23.2252	20.9226	18.6200	16.3174	14.0148	11.7122	9.4097	7.1075	4.8091	2.5474	.6114	.001841
4.7	32.4140	30.1114	27.8088	25.5062	23.2037	20.9011	18.5985	16.2959	13.9933	11.6907	9.3882	7.0860	4.7877	2.5268	.5979	.001635
4.8	32.3929	30.0904	27.7878	25.4852	23.1826	20.8800	18.5774	16.2748	13.9723	11.6697	9.3671	7.0650	4.7667	2.5068	.5848	.001453
4.9	32.3723	30.0697	27.7672	25.4646	23.1620	20.8594	18.5568	16.2542	13.9516	11.6491	9.3465	7.0444	4.7462	2.4871	.5721	.001291

Theis drawdown curve

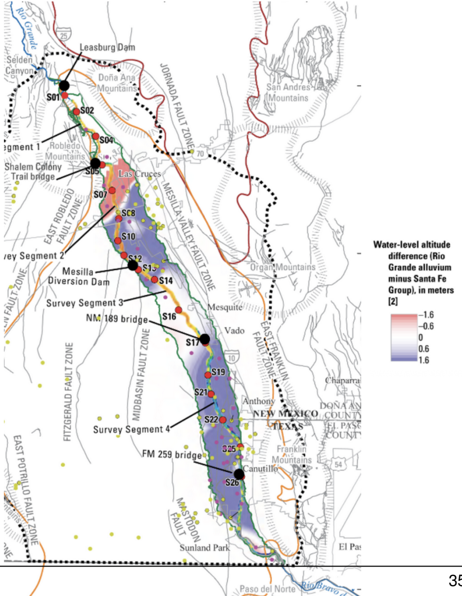




Rio Grande Stream Depletion

The Rio Grande is facing multiple factors leading to stream drying:

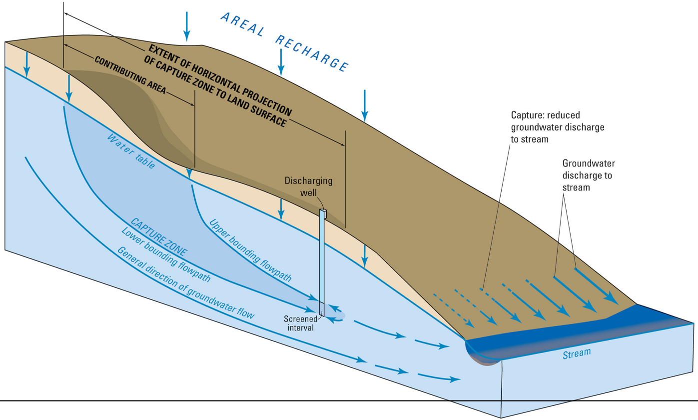
- Drought
- Climate Change
- Diversion
- Capture



Safe vs Sustainable Yield

Safe Yield: extraction should not exceed natural recharge to a system

Concept of capture means that pumping can draw from induced recharge and/or reduced discharge, so safe yield is not a good management metric.



Safe vs Sustainable Yield

Sustainable Yield: groundwater withdrawal rate that does not produce undesirable effects:

- stream depletion
- land subsidence
- seawater intrusion
- domestic wells drying
- WQ degradation
- ecosystem impacts

Groundwater Age-Sustainability Myth

"Fossil" water ($>$12,000 yrs) is overused and implies that aquifers with old water are not currently being replenished and are therefore being "mined."

Both old and young water ($<$50 yrs) can be used in sustainable and unsustainable ways (Ferguson et al. (2020).